

Lesson 6-10: Divide Mixed Numbers

Grade 6 Mathematics · Standard 6.NOS.1

Name: _____

Date: _____

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

Question 1 SELECTED RESPONSE · 1 POINT

What is $1\frac{1}{2} \div \frac{3}{4}$?

- ☐ A $1\frac{1}{8}$
- ☐ B 2
- ☐ C $\frac{3}{4}$
- ☐ D $\frac{9}{8}$

Question 2 SELECTED RESPONSE · 1 POINT

Convert $3\frac{2}{5}$ to an improper fraction.

- ☐ A $\frac{6}{5}$
- ☐ B $\frac{15}{5}$
- ☐ C $\frac{17}{5}$
- ☐ D $\frac{32}{5}$

Question 3 SELECTED RESPONSE · 1 POINT

What is $2\frac{1}{4} \div \frac{3}{4}$?

- ☐ A $1\frac{1}{2}$
- ☐ B $2\frac{1}{3}$
- ☐ C 3
- ☐ D $\frac{9}{16}$

Question 4

SELECTED RESPONSE · 1 POINT

What is $3\frac{1}{2} \div \frac{1}{4}$?

- ☐ A $3\frac{1}{8}$
- ☐ B $\frac{7}{8}$
- ☐ C $\frac{13}{4}$
- ☐ D 14

Question 5

SELECTED RESPONSE · 1 POINT

A baker has $2\frac{1}{4}$ cups of butter. Each batch of cookies needs $\frac{3}{4}$ cup of butter. How many batches can the baker make?

- ☐ A $1\frac{1}{2}$ batches
- ☐ B 2 batches
- ☐ C 3 batches
- ☐ D 6 batches

Question 6

SELECTED RESPONSE · 1 POINT

Mission 1: The stakeout shift is $4\frac{1}{2}$ hours long. Each agent covers a $1\frac{1}{2}$ -hour watch. How many agent watches fill one shift?

- ☐ A $2\frac{1}{2}$
- ☐ B 3
- ☐ C $4\frac{1}{2}$
- ☐ D 6

Part B — Test-Format Questions

These match the state test's formats: two-part questions, select-all, and written responses.

Question 7**TWO-PART QUESTION****Part A**

What is the value of $3\frac{3}{4} \div 1\frac{1}{2}$?

Fill in the bubble next to the one best answer.

- ☐ (A) $1\frac{1}{2}$
- ☐ (B) $2\frac{1}{2}$
- ☐ (C) 2
- ☐ (D) $5\frac{5}{8}$

Part B

Answer Part A before Part B — Part B asks about your reasoning.

Which reasoning shows why your answer to Part A is correct?

- ☐ (A) Convert $3\frac{3}{4}$ by multiplying 3×4 and keeping the denominator, giving $12/4$, then $12/4 \times 2/3 = 2$.
- ☐ (B) Ignore the whole numbers and divide only the fraction parts: $3/4 \div 1/2 = 3/4 \times 2/1 = 6/4 = 1\frac{1}{2}$.
- ☐ (C) Convert to improper fractions, $15/4$ and $3/2$, then keep, change, flip: $15/4 \times 2/3 = 30/12 = 5/2 = 2\frac{1}{2}$.
- ☐ (D) Convert to improper fractions and multiply straight across without flipping: $15/4 \times 3/2 = 45/8 = 5\frac{5}{8}$.

Question 8**WRITTEN RESPONSE · 2 POINTS****Type II Reasoning: Skipping the Conversion**

Tyrone divided $2\frac{1}{2} \div 1/4$. His work was: keep $2\frac{1}{2}$, change $\div$ to $\times$, flip $1/4$ to $4/1$. Then he multiplied only the whole number, $2 \times 4 = 8$, and carried the $1/2$ along, so he answered $8\frac{1}{2}$.

Explain Tyrone's misconception and give the correct quotient.

Write your answer in complete sentences. Show or explain your mathematical thinking.

Part C — Show Your Thinking

Answer in writing, the way the state test's constructed responses work.

Question 9 WRITTEN RESPONSE · 2 POINTS

A detective has a rope that is $5\frac{1}{4}$ feet long. She needs to cut it into pieces that are each $\frac{3}{4}$ foot for tying evidence tags. How many pieces can she cut? Write the equation, show each step (convert, keep-change-flip, multiply, simplify), and verify your answer.

Write your answer in complete sentences. Show or explain your mathematical thinking.

When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.